

COURSE HANDOUT		
1	Course details:	
	Faculty name	Dr. Sreearravind M
	Programme	B.E. – Mechanical Engineering
	Semester	03
	Section	01
	Course code	ME25C08
	Pre Requisite	–
	Course title	METALLURGY AND MATERIALS SCIENCE
2	Vision of the Department of Mechanical Engineering:	
	To look for greatness in the field of Mechanical Engineering training through information and abilities to cater to the necessities of society.	
3	Mission of the Department of Mechanical Engineering:	
	M1	To constantly improve the teaching and learning cycle to enable students to meet global needs.
	M2	To upgrade the knowledge and skills of students, faculty members and supporting staff through regular training.
	M3	To produce the best minds of engineers capable of meeting the expectations of industry, society and entrepreneurship development.
4	Programme Educational Objectives (PEOs):	
	PEO1	Effectuating success in careers by exploring with the design, digital and computational analysis of engineering systems, experimentation and testing, smart manufacturing, technical services, and research.
	PEO2	Amalgamating effectively with stakeholders to update and improve their core competencies and abilities to ethically compete in the ever-changing multicultural global enterprise.
	PEO3	To encourage multi-disciplinary research and development to foster advanced technology, and to nurture innovation and entrepreneurship in order to compete successfully in the global economy.
	PEO4	To globally share and apply technical knowledge to create new opportunities that proactively advances our society through team efforts and to solve various challenging technical, environmental and societal problems.
	PEO5	To create world class mechanical engineers capable of practice engineering ethically with a solid vision to become great leaders in academia, industries and society.
5	Programme Outcomes (POs):	
	PO1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
	PO2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
	PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.
	PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

	PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and communication tools including prediction and modelling to complex engineering activities with an understanding of the limitations.																																																																																																
	PO6	The engineer and society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.																																																																																																
	PO7	Environment and sustainability: Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for, sustainable development.																																																																																																
	PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.																																																																																																
	PO9	Individual and team work: Function effectively as an individual and as a member or leader in diverse teams, and in multidisciplinary settings.																																																																																																
	PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and society at large, including comprehending and writing effective reports and design documentation, making effective presentations, and giving and receiving clear instructions.																																																																																																
	PO11	Project management and finance: Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.																																																																																																
	PO12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.																																																																																																
6	Programme Specific Outcomes (PSOs) (if any):																																																																																																	
	PSO1	Apply the knowledge gained in Mechanical Engineering for design, development and manufacture of engineering systems.																																																																																																
	PSO2	Apply the knowledge acquired to investigate research-oriented problems in mechanical engineering with due consideration for environmental and social impacts.																																																																																																
	PSO3	Use engineering analysis and data-management tools for effective management of multidisciplinary projects.																																																																																																
7	Course Outcomes (COs):																																																																																																	
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	CO5	Explain the testing of mechanical properties and major deformation and failure mechanisms.																																																																																																
8	Correlation of COs with POs/PSOs																																																																																																	
		<table><tr><th>COs</th><th>PO1</th><th>PO2</th><th>PO3</th><th>PO4</th><th>PO5</th><th>PO6</th><th>PO7</th><th>PO8</th><th>PO9</th><th>PO10</th><th>PO11</th><th>PO12</th><th>PSO1</th><th>PSO2</th><th>PSO3</th></tr><tr><td>CO1</td><td>3</td><td>3</td><td>1</td><td>1</td><td></td><td>1</td><td>1</td><td></td><td></td><td>1</td><td></td><td>2</td><td>2</td><td>1</td><td></td></tr><tr><td>CO2</td><td>3</td><td>3</td><td>2</td><td>2</td><td>1</td><td></td><td>1</td><td></td><td></td><td>1</td><td></td><td>2</td><td>3</td><td>2</td><td>1</td></tr><tr><td>CO3</td><td>3</td><td>2</td><td>2</td><td>1</td><td></td><td>1</td><td>2</td><td></td><td></td><td>1</td><td></td><td>2</td><td>3</td><td>2</td><td></td></tr><tr><td>CO4</td><td>3</td><td>2</td><td>2</td><td>1</td><td>1</td><td>1</td><td>2</td><td></td><td></td><td>1</td><td></td><td>2</td><td>2</td><td>2</td><td>1</td></tr><tr><td>CO5</td><td>3</td><td>3</td><td>1</td><td>3</td><td>2</td><td>2</td><td>2</td><td>1</td><td>1</td><td>2</td><td>1</td><td>2</td><td>2</td><td>3</td><td>2</td></tr></table>	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	CO1	3	3	1	1		1	1			1		2	2	1		CO2	3	3	2	2	1		1			1		2	3	2	1	CO3	3	2	2	1		1	2			1		2	3	2		CO4	3	2	2	1	1	1	2			1		2	2	2	1	CO5	3	3	1	3	2	2	2	1	1	2	1	2	2	3	2
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3																																																																																			
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3–High; 2–Medium; 1–Low																																																																																																		

9	Evaluation Components
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	Evaluation Component	Duration	Marks (100)	Date & Time	Nature of Component	Evaluation Component
	Assessment Test–1	180 mins	100	As per the Academic Schedule/ Calendar provided by the University and college	Closed Book	Assessment Test–1
	Assessment Test–2	180 mins	100		Closed Book	Assessment Test–2
	Model Exam	180 mins	100		Closed Book	Model Exam
	# To conduct three internal assessment tests 1, 2 and 3 and one model examination, each with 100 marks. # The total marks obtained in the three assessment tests and one model examination shall be proportionately reduced to the internal-assessment mark prescribed by the institution and rounded to the nearest integer.					
10	List of teaching–learning pedagogy:					
	• Lesson plan and course handout • Session plan and session handout • Open hours • Students' seminar • Poster presentation • Content beyond the syllabus • Learning Management System (LMS) • Collaborative teaching/learning–joint presentation or group problem solving • Assignments / Quiz					
11	Open hour for students:					
	• Tuesday (12.40 p.m. to 01.30 p.m.)					
12	Link address for course materials:					
	1. https://nptel.ac.in/courses/113102080 2. https://nptel.ac.in/courses/113104068					
13	Detailed academic calendar of lecture topics (date, topic, learning outcomes of each topic, related unit, total number of lectures in a unit and book/chapter to be read for the topic)					
	Appendix 4.1					
14	Entire syllabus					
	Appendix 4.2					
15	Recommended list of e-books:					
	1. William D. Callister Jr. and David G. Rethwisch, <i>Materials Science and Engineering: An Introduction</i> , Wiley. 2. Kenneth G. Budinski and Michael K. Budinski, <i>Engineering Materials</i> , 9th ed., Prentice Hall of India, 2018. 3. Sydney H. Avner, <i>Introduction to Physical Metallurgy</i> , McGraw-Hill, 1994.					
16	Recommended list of online courses such as SWAYAM/NPTEL/MOOCs:					
	For NPTEL online courses: • Introduction to Materials Science and Engineering: https://nptel.ac.in/courses/113102080 • Phase Diagrams in Materials Science and Engineering: https://nptel.ac.in/courses/113104068					
17	Content beyond Syllabus:					
	• Data-driven materials selection and introductory materials informatics • Sustainable materials selection, recycling and circular-material flows • Additive-manufacturing process–microstructure–property relationships • Failure-analysis documentation and digital traceability of test data					
18	Recommended list of mini projects/projects/technical training etc.:					

	• Phase-diagram-based alloy and heat-treatment selection for a component • Comparative study of ferrous, non-ferrous, polymer, ceramic and composite materials • Hardness and tensile-property data analysis for engineering materials • Failure-mode identification using representative fracture-surface case studies
19	Students' Presentation:
	• Materials selection for automotive, aerospace and energy applications • Nickel-based superalloys and shape-memory alloys • Engineering polymers, ceramics and composites in modern products • Industrial fatigue, creep and fracture case studies
20	List of technical trainings
	• Metallographic sample preparation and microstructure interpretation • Tensile, impact and hardness testing procedures • Introductory phase-diagram and materials-selection software • Engineering materials standards and test-report documentation
21	Additional information
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Appendix 4.1
Detailed academic calendar of lecture topics

Lecture No.	Date	Topics to be covered	Learning outcomes of each topic	Teaching Methodology	Revised Bloom's Taxonomy level	Periods	Cumulative Periods	References
UNIT-I: CONSTITUTION OF ALLOYS AND PHASE DIAGRAMS								
1	01-07-26	Constitution of alloys and classification of engineering alloys	Explain the constitution and basic classification of alloys.	BB	Understand	1	1	T1
2	02-07-26	Solid solutions: substitutional and interstitial	Distinguish substitutional and interstitial solid solutions with examples.	BB	Understand	1	2	T1
3	03-07-26	Binary phase diagrams: basic terms and interpretation	Identify phases, phase fields, tie lines and phase boundaries in binary diagrams.	BB	Remember / Understand	1	3	T2
4	06-07-26	Isomorphous phase diagram	Interpret an isomorphous phase diagram and apply the lever rule.	BB / PS	Apply	1	4	T2
5	08-07-26	Eutectic and eutectoid reactions	Explain eutectic and eutectoid reactions and the resulting microstructures.	BB	Understand	1	5	T1
6	10-07-26	Peritectic and peritectoid reactions	Explain peritectic and peritectoid reactions using schematic diagrams.	BB	Understand	1	6	T1
7	13-07-26	Iron-iron carbide equilibrium diagram	Interpret the Fe-Fe ₃ C diagram, phases and invariant reactions.	BB / PS	Apply / Analyse	1	7	T2
8	15-07-26	Classification of steels: microstructure, properties and applications	Classify steels and relate carbon content to microstructure and properties.	BB	Analyse	1	8	T2
9	17-07-26	Classification of cast irons: microstructure, properties and applications	Compare grey, white, malleable and spheroidal graphite cast irons.	BB	Analyse	1	9	T1
Assignment I								

Appendix 4.1 (Continued)

Lecture No.	Date	Topics to be covered	Learning outcomes of each topic	Teaching Methodology	Revised Bloom's Taxonomy level	Periods	Cumulative Periods	References
UNIT-II: HEAT TREATMENT								
10	20-07-26	Heat treatment: definition, objectives and process variables	Explain the purpose and principal variables of heat treatment.	BB	Understand	1	10	T2
11	22-07-26	Full annealing and stress-relief annealing	Explain the cycles, microstructural changes and applications of annealing treatments.	BB	Understand	1	11	T1
12	23-07-26	Recrystallisation and spheroidising annealing	Differentiate recrystallisation and spheroidising treatments.	BB	Understand	1	12	T1
13	25-07-26	Normalising, hardening and tempering of steel	Select normalising, hardening and tempering cycles for a given requirement.	BB / PS	Apply	1	13	T2
14	27-07-26	Isothermal transformation (TTT) diagrams	Interpret TTT diagrams and predict transformation products.	BB / PS	Apply / Analyse	1	14	T1
15	29-07-26	Continuous cooling transformation (CCT) diagrams	Compare TTT and CCT diagrams for practical heat-treatment conditions.	BB / PS	Analyse	1	15	T1
16	31-07-26	Austempering and martempering	Explain austempering and martempering cycles and their advantages.	BB	Understand	1	16	T2
17	03-08-26	Hardenability and Jominy end-quench test	Evaluate hardenability using the Jominy end-quench test.	BB / PS	Apply / Analyse	1	17	T2
18	05-08-26	Case and surface hardening; thermo-mechanical treatment and sintering	Compare carburising, nitriding, cyaniding, carbonitriding, flame, induction, vacuum and plasma hardening; outline thermo-mechanical treatment and sintering.	BB	Understand / Analyse	1	18	T1
Internal Assessment Test 1: 10-08-26 to 17-08-26 (No classes)								

Appendix 4.1 (Continued)

Lecture No.	Date	Topics to be covered	Learning outcomes of each topic	Teaching Methodology	Revised Bloom's Taxonomy level	Periods	Cumulative Periods	References
UNIT-III: FERROUS AND NON-FERROUS METALS								
19	06-08-26	Effect of alloying elements on steels: Mn, Si, Cr and Mo	Explain how Mn, Si, Cr and Mo modify steel properties and performance.	BB	Understand	1	19	T2
20	07-08-26	Effect of alloying elements on steels: Ni, V, Ti and W	Explain how Ni, V, Ti and W influence microstructure and properties.	BB	Understand	1	20	T1
21	08-08-26	Stainless steels and tool steels	Classify stainless and tool steels and select them for engineering applications.	BB	Apply	1	21	T2
22	18-08-26	HSLA and maraging steels	Compare HSLA and maraging steels in terms of strengthening and applications.	BB	Analyse	1	22	T1
23	19-08-26	Alloy cast irons	Explain the microstructure, properties and applications of alloy cast irons.	BB	Understand	1	23	T1
24	21-08-26	Copper and its alloys: brass, bronze and cupronickel	Differentiate major copper alloys and select them for service conditions.	BB	Apply	1	24	T2
25	24-08-26	Aluminium alloys and Al-Cu precipitation strengthening	Explain aluminium alloy classification and the stages of precipitation strengthening.	BB / PS	Apply / Analyse	1	25	T2
26	27-08-26	Titanium alloys, magnesium alloys and nickel-based superalloys	Compare Ti, Mg and Ni-based alloys based on properties and applications.	BB	Analyse	1	26	T1
27	31-08-26	Shape-memory alloys and overview of materials standards	Explain shape-memory behaviour and the role of materials standards in engineering practice.	BB / PPT	Understand / Apply	1	27	R4
Assignment II								

Appendix 4.1 (Continued)

Lecture No.	Date	Topics to be covered	Learning outcomes of each topic	Teaching Methodology	Revised Bloom's Taxonomy level	Periods	Cumulative Periods	References
UNIT-IV: NON-METALLIC MATERIALS								
28	01-09-26	Polymers: classification, structure and engineering significance	Classify polymers and distinguish commodity and engineering polymers.	BB	Understand	1	28	T1
29	02-09-26	Properties and applications of PE, PP, PS and PVC	Compare common commodity polymers and their engineering applications.	BB	Understand	1	29	T1
30	03-09-26	Properties and applications of PMMA, PET, PC, PA and ABS	Select common engineering polymers based on their property profiles.	BB	Apply	1	30	R3
31	07-09-26	High-performance polymers: PAI, PPO, PPS, PEEK and PTFE	Explain the advantages and limitations of high-performance polymers.	BB	Understand / Analyse	1	31	R3
32	09-09-26	Thermoset polymers: urea-formaldehyde and phenol-formaldehyde	Differentiate thermosets from thermoplastics and describe their applications.	BB	Understand	1	32	T1
33	11-09-26	Engineering ceramics: Al ₂ O ₃ , SiC and Si ₃ N ₄	Relate ceramic structure and properties to engineering applications.	BB	Analyse	1	33	T1
34	15-09-26	PSZ, SIALON and inter-metallic materials	Explain the properties and applications of PSZ, SIALON and intermetallics.	BB	Understand	1	34	R5
35	18-09-26	Composites: matrix and reinforcement materials	Classify composite constituents and explain load transfer between matrix and reinforcement.	BB	Analyse	1	35	T1
36	21-09-26	Applications of composites and nanocomposites	Select composites and nanocomposites for representative engineering applications.	BB / PPT	Apply	1	36	R5
Internal Assessment Test 2: 29-09-26 to 06-10-26 (No classes)								

Appendix 4.1 (Continued)

Lecture No.	Date	Topics to be covered	Learning outcomes of each topic	Teaching Methodology	Revised Bloom's Taxonomy level	Periods	Cumulative Periods	References
UNIT-V: MECHANICAL PROPERTIES AND DEFORMATION MECHANISMS								
37	22-09-26	Mechanisms of plastic deformation: slip	Explain slip systems and the role of dislocations in plastic deformation.	BB	Understand	1	37	T2
38	23-09-26	Twinning and comparison with slip	Differentiate twinning from slip and identify conditions favouring each mechanism.	BB	Analyse	1	38	T2
39	24-09-26	Types of fracture: ductile, brittle and fatigue fracture	Classify fracture modes using macroscopic and microscopic features.	BB / PPT	Analyse	1	39	T1
40	25-09-26	Fracture mechanics and Griffith theory	Apply the Griffith energy criterion to ideal brittle fracture.	BB / PS	Apply / Analyse	1	40	R4
41	26-09-26	Testing under tensile, compression and shear loads	Explain test procedures and interpret basic stress-strain responses.	BB / PS	Apply	1	41	T1
42	07-10-26	Hardness tests: Brinell, Vickers and Rockwell	Compare hardness-test principles, scales and applications.	BB / Demo	Apply / Analyse	1	42	T1
43	08-10-26	Micro-hardness and nano-hardness tests	Explain indentation-size effects and the use of micro- and nano-hardness testing.	BB / PPT	Understand	1	43	R5
44	10-10-26	Impact tests: Izod and Charpy	Differentiate Izod and Charpy tests and interpret impact-energy results.	BB / Demo	Apply / Analyse	1	44	T1
45	13-10-26	Fatigue and creep failure mechanisms	Explain fatigue and creep mechanisms, characteristic curves and engineering significance.	BB	Analyse	1	45	T1
46	15-10-26 16-10-26	Presentations on technical applications of engineering materials	Present and discuss a materials-selection or failure-analysis case study.	PPT	Understand / Analyse	2	47	
Internal Assessment Test 3								
Model Examination								

Note:

BB – Black Board FC – Flipped Classroom
PPT – Power Point Presentation PS – Problem Solving
Demo – Demonstration T – Text Book
R – Reference Book

Appendix 4.2

Catalog Description

This course introduces the constitution, processing, properties and engineering applications of metallic and non-metallic materials. It covers alloy constitution and phase diagrams, the iron–iron carbide equilibrium diagram, heat-treatment processes, ferrous and non-ferrous alloys, polymers, ceramics and composites, and the mechanical testing and failure mechanisms of engineering materials. The course emphasises the relationship among processing, microstructure, properties and performance for informed materials selection.

Course Objectives

1. To learn the construction and interpretation of phase diagrams and the use of the iron–iron carbide phase diagram for understanding microstructure formation.
2. To learn the selection and application of heat-treatment processes and the associated microstructure formation.
3. To illustrate different ferrous and non-ferrous alloys and their uses in engineering.
4. To illustrate polymers, ceramics and composites and their uses in engineering.
5. To learn material-testing procedures and failure mechanisms in engineering applications.

Text Books

1. Kenneth G. Budinski and Michael K. Budinski, “Engineering Materials,” Prentice Hall of India Private Limited, 9th edition, 2018.
2. Sydney H. Avner, “Introduction to Physical Metallurgy,” McGraw-Hill Book Company, 1994.

Reference Books

1. A. Alavudeen, N. Venkateshwaran and J. T. Winowlin Jappes, *A Textbook of Engineering Materials and Metallurgy*, Laxmi Publications, 2006.
2. Amandeep Singh Wadhwa and Harvinder Singh Dhaliwal, *A Textbook of Engineering Material and Metallurgy*, University Sciences Press, 2008.
3. G. S. Upadhyay and Anish Upadhyay, “Materials Science and Engineering,” Viva Books Pvt. Ltd., New Delhi, 2020.
4. V. Raghavan, “Materials Science and Engineering,” Prentice Hall of India Pvt. Ltd., 6th edition, 2019.
5. William D. Callister, “Material Science and Engineering,” Wiley India Pvt. Ltd., 2nd edition reprint, 2019.

Course Content

UNIT-I: CONSTITUTION OF ALLOYS AND PHASE DIAGRAMS

Constitution of alloys – solid solutions, substitutional and interstitial; phase diagrams; isomorphous, eutectic, eutectoid, peritectic and peritectoid reactions; iron–iron carbide equilibrium diagram; classification of steel and cast iron; microstructure, properties and applications.

UNIT-II: HEAT TREATMENT

Definition – full annealing, stress relief, recrystallisation and spheroidising; normalising, hardening and tempering of steel; isothermal transformation diagrams; cooling curves superimposed on I.T. diagrams; continuous cooling transformation (CCT) diagrams; austempering, martempering, hardenability and Jominy end-quench test; case hardening, carburising, nitriding, cyaniding and carbonitriding; flame and induction hardening; vacuum and plasma hardening; thermo-mechanical treatments; elementary ideas on sintering.

UNIT-III: FERROUS AND NON-FERROUS METALS

Effect of alloying additions on steel (Mn, Si, Cr, Mo, Ni, V, Ti and W); stainless and tool steels; HSLA and maraging steels; grey, white, malleable, spheroidal and alloy cast irons; copper and its alloys–brass, bronze and cupronickel; aluminium and its alloys; Al–Cu precipitation-strengthening treatment; titanium alloys, magnesium alloys, nickel-based superalloys and shape-memory alloys; properties and applications; overview of materials standards.

UNIT-IV: NON-METALLIC MATERIALS

Polymers–types of polymers, commodity and engineering polymers; properties and applications of PE, PP, PS, PVC, PMMA, PET, PC, PA, ABS, PAI, PPO, PPS, PEEK and PTFE; thermoset polymers–urea- and phenol-formaldehydes; nylon; engineering ceramics–properties and applications of Al_2O_3 , SiC, Si_3N_4 , PSZ and SIALON; intermetallics; composites–matrix and reinforcement materials; applications of composites and nanocomposites.

UNIT-V: MECHANICAL PROPERTIES AND DEFORMATION MECHANISMS

Mechanisms of plastic deformation, slip and twinning; types of fracture; fracture mechanics; Griffith's theory; testing of materials under tensile, compression and shear loads; hardness tests–Brinell, Vickers and Rockwell; micro- and nano-hardness tests; impact tests–Izod and Charpy; fatigue and creep failure mechanisms.

TOTAL: 45 PERIODS (9 periods each)

OUTCOMES:

At the end of the course, the students would be able to:

1. Explain alloys and phase diagrams, the iron–iron carbide diagram and steel classification.
2. Explain isothermal transformation, continuous cooling diagrams and different heat-treatment processes.
3. Clarify the effect of alloying elements on ferrous and non-ferrous metals.
4. Summarize the properties and applications of non-metallic materials.
5. Explain the testing of mechanical properties and major deformation and failure mechanisms.

Appendix 4.3: Compliance report

Department of Mechanical Engineering	
Programme	B.E. – Mechanical Engineering
Head of the Dept	Dr. N. KEERTHIVASAN M.Tech., Ph.D.

Compliance report of course handout

Sl. No.	Course code	Course title	Section	Taught by faculty	Course handout submission date	Remarks by HoD, if any
1	ME25C08	Metallurgy and Materials Science	01	Dr. Sreearavind M		

Signature of Faculty

Signature of HOD